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Studierichting: Informatica

Oefeningenreeks 4A nr. 17

$$\int \sin x \cos x dx$$

17.1)

$$= \int \sin x d(\sin x) \stackrel{t=\sin x}{=} \int t dt = \frac{1}{2} t^2 \stackrel{t=\sin x}{=} \frac{1}{2} \sin^2 x + c$$

17.2)

$$= - \int \cos x d(\cos x) \stackrel{t=\cos x}{=} - \int t dt = -\frac{1}{2} t^2 \stackrel{t=\cos x}{=} -\frac{1}{2} \cos^2 x + c$$

17.3)

$$= \frac{1}{2} \int \sin(2x) dx = \frac{1}{4} \int \sin(2x) d(2x) = -\frac{1}{4} \cos(2x) + c$$

References